Contents

modal_sound.py -- Your Toolkit for HW2

The Modal Synthesis Formula (van den Doel & Pai 1998)

Four Analytical Models

Rayleigh Damping

Material Presets

Quick Start (5 lines)

HW2 Workflow: Houdini → WAV

API at a Glance

Try It Now

`modal_sound.py` -- Your Toolkit for HW2

The Modal Synthesis Formula (van den Doel & Pai 1998)

$$y_k(t) = \sum_n ; a_{nk} ; e^{-d_n t} ; \sin(2\pi f_n, t)$$

- f_n -- modal frequencies (Hz) d_n -- decay rates (Rayleigh damping) a_{nk} -- mode-shape gains at strike point k

Four Analytical Models

Model	Geometry	Frequencies	Mode shapes
String	Fixed-fixed, 1D	$f_n = n, f_1$ (harmonic)	$\sin(n\pi x/L)$
Beam	Free-free, 1D (Euler-Bernoulli)	$f_n = f_1(\lambda_n/\lambda_1)^2$ (inharmonic)	Cosh-cos-sinh-sin combination
Membrane	Fixed edge, circular	$f_{mn} = f_1, z_{mn}/z_{01}$ (Bessel zeros)	$J_m(z_{mn} r/a)\cos(m\theta)$
Plate	Simply supported, rectangular	$f_{mn} = f_1(m^2 + n^2R^2)/(1 + R^2)$	$\sin(m\pi x/a)\sin(n\pi y/b)$

Rayleigh Damping

$$d_n = \frac{\alpha}{2} + \frac{\beta_i \omega_n^2}{2}, \qquad \omega_n = 2\pi f_n$$

α (mass) = uniform decay | β (stiffness) = damps high frequencies faster

Material Presets

Material	f_1 (Hz)	α (1/s)	β (s)	Character
Steel	440	1	2e-6	Long ring, bright
Glass	880	0.5	1e-6	High, clear ring
Wood	220	40	5e-5	Short, warm thud
Ceramic	660	5	1e-5	Mid-ring, dense
Rubber	110	200	1e-3	Dead thud

Quick Start (5 lines)

```
import modal_sound as ms

model = ms.make_model("plate", "glass", num_modes=20,
                      strike_positions=(0.5, 0.5))
audio = ms.synthesize_impact(model, duration=3.0)
ms.write_wav("glass_plate.wav", audio)
```

HW2 Workflow: Houdini → WAV

```
model  = ms.make_model("plate", "steel", num_modes=24,
                       f1_override=120.0,          # tune to your object
                       strike_positions=positions)  # pre-sample K points
```

```
events = ms.load_contact_events("houdini_impacts.csv")
audio  = ms.synthesize_contact_events(model, events)
ms.write_wav("my_sound.wav", audio)
```

- Export contact **time**, **amplitude**, **position** from Houdini as CSV
- Override f_1 to match your object's size; keep α, β from a material preset
- `strike_positions` controls spatial variation -- hitting a node silences that mode

API at a Glance

Function	What it does
`make_model(shape, material, ...)`	Build a `ModalModel` from shape + preset
`make_string/beam/membrane/plate_model(...)`	Build with explicit f_1, α, β
`synthesize_impact(model, ...)`	Single impact → audio array
`synthesize_contact_events(model, events)`	Event sequence → audio array
`load_contact_events(csv_path)`	CSV → event list
`write_wav(path, audio)`	Audio array → 16-bit WAV
`compute_t60(model)`	60 dB decay time: $T_{60} = 6 \ln 10; /; d_n$

Try It Now

```
python modal_sound.py
```

Generates demo WAVs for all four object types -- listen and compare.

Also see: **Modal Sound Explorer** demo at https://dougjam.github.io/demos/modal-sound-explorer/